

Samuele TONDELLI

Computer Science and Engineering Student

 [linkedin.com/in/samuele-tondelli-681691356](https://www.linkedin.com/in/samuele-tondelli-681691356)

 github.com/SamueleTondelli

 tondellisamuele1@gmail.com

 [Personal website](#)

“ SUMMARY

Final-year master’s student in Computer Science and Engineering focused on ML systems, GPU computing, and performance engineering. Currently optimizing GPU inference of a graph transformer for protein docking. Background in low-level systems programming, FPGA acceleration, and embedded software. Seeking internship and research opportunities.

EDUCATION

2024 - Present	Master in Computer Science and Engineering, Politecnico di Milano , expected to graduate in July 2027
2021 - 2024	Bachelor in Computer Engineering, University of Modena and Reggio Emilia . Thesis : "Optimization and synthesis on FPGA of the logarithm of softmax function for neuromorphic accelerators". 110 cum laude

SKILLS

Languages	C, C++, Python, Rust, Assembly, Go, Java
GPU & performance	CUDA, SIMD, Triton, AVX-512, TensorRT, NSight Sytems/Compute
ML frameworks	PyTorch, Scikit-learn, Hydra, TensorFlow
Hardware & Embedded	Vitis HLS, STM32, Verilog, Xilinx Zynq FPGAs, Raspberry Pi
Dev tools	Linux, Git, Make, CMake, GCC, GDB, Valgrind, Vivado, ModelSim

EXPERIENCE

Feb 2026	Student Researcher Genomic ML, NECSTLAB, POLITECNICO DI MILANO, Italy
Sep 2025	➢ Conducting survey study on ML approaches for single nucleotide variant classification. ➢ Benchmarked 4 models (MLP, XGBoost, Logistic Reg, Random Forest) across 144 genomic datasets. ➢ Reduced training resource usage up to 10x while minimally impacting test performance. ➢ Analyzed pathogenicity on variants of uncertain significance (90% consistency across 72 models). ➢ Paper accepted at IEEE EMBC 2026 (Paper No. 3391).
	MLGenomicsPythonHydraScikit-learn
Sep 2025	Student Researcher FPGA Deep Space, NECSTLAB, POLITECNICO DI MILANO, Italy
Mar 2025	➢ Accelerated ECOS library for autonomous navigation of deep space satellites. ➢ Developed accelerator for sparse matrix computations (SpMV and LDLt) using Vitis HLS and Vivado. ➢ Deployed IP on Kria KD240 and developed a C Linux driver, achieving 2x speedup over software. ➢ Completed in the 2025 edition of the AMD Open Hardware competition.
	FPGAHLSVitisVivadoXilinxEmbedded systemsC
May 2024	Software Engineer Intern Formula SAE, MORE MODENA RACING, UNIMORE, Italy
Mar 2023	➢ Co-led a team of 4 developing software for combustion and hybrid Formula SAE cars. ➢ Developed live, wireless telemetry (1 Mbps over 500m) using Raspberry Pi 5 and Grafana. ➢ Developed car dashboard software on STM32 controller reading CAN bus data.
	AutomotiveCRaspberry PiLive telemetryGrafanaSTM32CAN

PROJECTS

GPU ACCELERATED GRAPH TRANSFORMER INFERENCE	2026
Master’s thesis Politecnico di Milano, private repository	
Optimizing GPU inference of a graph transformer for protein docking, in active development.	
➢ Implemented LibTorch C++ and TensorRT backends running on an NVIDIA A100. ➢ Profiled with NSight Systems and removed host-device synchronization, reducing forward pass latency by 5x. ➢ Implemented custom Triton kernels for fused scatter-adds, with a 2x speedup over Torch baseline.	
GPUCUDA_TritonPyTorchTensorRTGraph TransformerNSight	
GRADINO - OPTIMIZED C AUTOGRAD ENGINE	2026
 github.com/SamueleTondelli/gradino	
Autograd engine written in C optimized for CPU training with SIMD acceleration and custom allocators.	
CDeep learningSIMDAVX-512Arena allocators	

LANGUAGES

Italian	●●●●●
English	●●●●○

+ STRENGTHS

- Curiosity
- Motivation
- Passion